

## License plates

In the future, license plates will be supplemented with RFID chips containing information about the car and its owners

## RFID chips

Electronic toll collection, using smart cards and RFID chips has been implemented along the Delhi-Gurgaon highway—cars don't need to stop to pay toll

which look more like floating silver carpets. Solar cells are bound to get more efficient, and we may one day see a car that runs purely on solar energy. However, for now they are destined only to be conceptual contraptions.

### Cognizant cars

Three out of every five drivers don't know how to drive properly, and the cars don't seem to mind. A day will come when cars, like children growing up, will realize that running into a tree might be counterproductive. Tomorrow's cars will need to be smart, maybe not as smart as KITT, but smart enough to know when more brakes are required than applied by the driver in-charge. Today's cars have as many as 50 microprocessors controlling various aspects of their workings – from managing the car's AC temperature and stereo volume, to controlling the rate of firing of the spark plugs. Throw in a couple of hundred more, and the car can be made aware of its location via GPS and aware of any surrounding cars, aware of traffic signals, and aware of the fact that the idiot behind the wheel is just too drunk to drive.

### Braking assist

A large number of accidents are caused due to inadequate braking, when it is needed the most. Brake-assist systems can greatly decrease the chances of an accident, and otherwise decrease the impact of a collision. Although collapsible steering wells and air-bags assuage the blow when the accident happens, this technology takes the proactive step of decreasing the chance of an accident ever happening. Systems developed by Volvo and Daimler-Benz are intelligent enough to understand when the brakes are being applied in an emergency and will boost the power to the brakes so that the car stops in time. Volvo's CWAB (Collision Warning with Auto Brake) system has radar to detect possible collisions, and appropriately power up the brakes, so that sufficient braking is achieved. It can even respond with warning lights and sounds if the driver fails to respond in time, and will automatically brake to decrease collision speed even after the collision has become unavoidable.

### Cruise control

Hopefully, by the time this technology hits mainstream cars, we would have forgotten about *Speed II: Cruise Control*, to not be put off by its name. Cars with Autonomous Cruise Control are capable of anything from automatically controlling the car's speed to completely unassisted driving. This one seems straight out of science fiction, but in fact is already available, and had been demonstrated as far back as 1995 by Mercedes-Benz with a specially designed S-Class car. Using lasers or radar, cars with Autonomous Cruise Control systems can locate and track other traffic on the streets, and appropriately change the speed of the car – slowing down in traffic and speeding up when the roads clear. They also use information about speed limits from an in-built GPS to ensure that the car always stays within driving limits.

Best of all, this is available today! If you're willing to shell out a few crores of course. Do not despair; over time these features will trickle down to the cars most people can afford, just like ABS (Anti-lock Braking System) did. The future is here! Really, we checked.

### Assisted parking

How many times have you nicked someone's bumper while trying to parallel park? Afraid someone will catch you in the act one of these days? Assisted parking means never having to say sorry; it's like love, but less complicated. With assisted

parking systems, all you need to do is align the car near the parking position, and tell it where to park, and the car will park itself. Is this even important enough to implement? Why not park yourself? Parking is one of those few instances where we often require the car to go backwards, occasionally requiring third-party assistance in assuring us we are doing it right. A lot of time is wasted in parking procedures, and if this repetitive task can be left to the car,

so much the better. This may not seem essential, but

it surely is beneficial.

Current technologies aren't smart enough to locate a parking spot, and park automatically, they need some input from the user as to where to park. Fully-automated parking will surely be available someday, and society will be left with one less nuisance to joke about.

With the parking left to the car, the driving to the cruise control, and with braking assist to make ensure that you won't bash into anyone, the future will be a much safer place to drive. Who knows, passengers might even be recommended to pass out drunk before driving, ensuring that they don't mess with the controls?!

### The shape of things to come

Now that we've paid attention to the inner beauty of cars to come, we can safely discuss that which we really care about the most – its looks! The fact is, looks matter. No matter how much you crave a car with muscle, you'd

### NUCLEAR CARS!

OK this isn't really available, but back in 1958, when the word 'nuclear' was still cool, Ford unveiled a concept design of a car which would feature a Nuclear reactor for running the car. It never went into production, and was in fact never even built. Apparently, giving everyone a small nuclear reactor is a bad idea.

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